

Rossmann Sales Forecasting

Comprehensive Analytical Report & Market Dynamics Study

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EXECUTIVE SUMMARY

Rossmann Pharmaceuticals operates a vast network of retail stores across multiple cities and requires an accurate, highly scalable method to forecast daily sales up to six weeks in advance. Reliable sales projections serve as the foundational bedrock for capital allocation, inventory management, supply chain optimization, staffing schedules, and targeted promotional campaigns.

Historically, store managers relied on personal judgment and localized intuition. While valuable, this decentralization resulted in notable forecasting inconsistencies and operational friction. This report establishes an end-to-end data processing, validation, and exploratory framework designed to support machine learning models capable of delivering precise store-level sales forecasts.

TOTAL HISTORICAL TURNOVER	AVG DAILY SALES / STORE	SALES PER CUSTOMER	PROMO-DAY SHARE
€5.87B	€6,572	€9.49	44.6%
Across 1,115 Retail Stores	Open store-day basis	Average basket purchasing size	Percentage of active store days

BUSINESS & ANALYTICAL OBJECTIVES

The core objective is to construct an analytical pipeline that ingests raw transactional records, cleans and validates structural integrity, enriches data with external and temporal parameters, and models daily sales volume across all store locations. Key variables evaluated include:

- Promotional Campaigns:** Daily store promotions and multi-year continuous Promo2 programs.
- Calendar & Holiday Effects:** State holidays (Public, Easter, Christmas) and state-level school holidays.
- Store Infrastructure:** Store classification types (a, b, c, d), assortment levels, and operational opening status.
- Competitive Landscape:** Proximity distance to nearest competitors and open-since dates.

DATA GOVERNANCE & QUALITY ASSESSMENT

To guarantee model robustness, an initial profiling of the raw datasets (`train.csv`, `test.csv`, `store.csv`) was conducted. Data integrity rules were rigorously tested to detect duplicates, invalid values, or logical conflicts.

Dataset	Total Records	Feature Count	Missing Values	Duplicate Keys	Temporal Span
Training Data	1,017,209	9	0	0	2013-01-01 to 2015-07-31
Test Data	41,088	8	11 (Open)	0	2015-08-01 to 2015-09-17
Store Master	1,115	10	1,801	0	N/A

Business Logic Validation Summary

All transactional records successfully passed critical business rule validations: zero negative sales or customer counts were detected, no closed store reported positive sales, and binary flags for opening status

and promotions were fully compliant. Missing values in competition distance and Promo2 details were treated structurally rather than deleted, preserving critical economic signals.

PROMOTION INTELLIGENCE & DISTRIBUTION ALIGNMENT

A key analytical finding is the comparative evaluation of promotional activity between historical training data and the six-week test forecasting period. Machine learning models require that key driver distributions remain stationary between training and inference environments.

Environment / Horizon	No Promo Share	Promo Share	Distribution Variance
Historical Training	61.85%	38.15%	Baseline
Future Test Period	62.00%	38.00%	-0.15% (Highly Similar)

Key Takeaway: The promo share between train and test datasets is remarkably consistent, exhibiting less than a 0.2 percentage point variation. This confirms there is no meaningful promotional distribution shift, allowing predictive models trained on historical data to generalize reliably without synthetic reweighting.

HOLIDAY EVENT STUDY & PURCHASING BEHAVIOR

Major state holidays (Public Holidays, Easter, Christmas) severely alter normal shopping patterns. An event study was constructed around a **14-day window** (–7 days prior to +7 days post-holiday) to quantify customer behavior before, during, and after major holiday events.

Relative Phase	Sales Index (Base=100)	Customer Traffic Index	Store Open Rate (%)	Avg Basket Value
Pre-Holiday (Days -7 to -1)	100.0	100.0	84.2%	€9.42
Holiday Event (Day 0)	3.1	2.8	0.6%	€10.85
Post-Holiday (Days +1 to +7)	91.4	90.2	82.5%	€9.55

Behavioral Dynamics:

- **Pre-Holiday Surge:** Days –2 and –1 experience an extreme uptick in customer volume and total sales (+32% over normal baseline) as households stockpile pharmaceutical and personal care essentials.
- **Holiday Closure:** Store open rates collapse to 0.6% on the holiday itself, reflecting mandatory legal closures. On-duty open stores enjoy significantly elevated average basket sizes (+15.2%).
- **Post-Holiday Slump:** Sales exhibit a minor hangover period during Days +1 to +3, operating at approximately 91.4% of pre-holiday baseline strength before normalizing by Day +5.

STRATEGIC RECOMMENDATIONS

1. **Inventory Buffer Pre-positioning:** Supply chain managers must increase inventory holding by 25–30% during the 5-day pre-holiday window to avoid stockouts on fast-moving consumer goods.
2. **Dynamic Staff Allocation:** Shift scheduling should mirror the pre-holiday demand surge, reallocating staff hours from post-holiday slump periods to peak pre-event days.

3. **Integrated Feature Pipeline:** Predictive ML models must explicitly feature relative-day lag and lead indicators around state holidays to properly capture non-linear demand shifts.